

Linking the Gut to the Brain

Can changing the gut microbiome help prevent Alzheimer's disease?

By Kenisya Saravenen | Biomedical Sciences | Neuroscience & Microbiology | 5-min read

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A disease conversation is moving beyond the brain

For decades, Alzheimer's research has focused heavily on processes inside the brain, particularly amyloid-beta accumulation, tau pathology and neuroinflammation. However, a different organ is increasingly entering the conversation: the gut. The gut-brain axis describes a two-way communication network linking the gastrointestinal tract and central nervous system through microbial metabolites, immune signalling and neural and endocrine pathways.[1]

That shift has created a provocative biomedical question. Could an altered gut microbiome influence the conditions that accompany neurodegeneration and if so, could changing the microbiome become part of dementia prevention? The evidence is intriguing, but it is not yet strong enough to support the idea of a microbiome "cure" for Alzheimer's disease.

From association to mechanism

One of the main clues is gut dysbiosis: changes in the composition or function of intestinal microorganisms. Human studies have reported differences in microbial communities among people with cognitive impairment and Alzheimer's disease, including changes in diversity and in bacteria involved in producing short-chain fatty acids (SCFAs). A 2026 systematic review of 15 intervention studies involving 4,275 participants found that microbiota-targeted interventions were associated with improvements in some cognitive outcomes, particularly in people with mild or prodromal impairment, while also highlighting major heterogeneity and the need for larger, longer randomized trials.[2]

SCFAs such as acetate, propionate and butyrate are especially interesting because they can act as signaling molecules between microbes and host tissues. They have been investigated for roles in intestinal barrier function, immune regulation and inflammatory signaling. The proposed pathway is therefore more complex than the slogan "bad bacteria cause Alzheimer's": diet can alter microbial metabolism, microbial products can alter host signaling, and those signals may influence the wider inflammatory environment.[1,2]

What does the experimental evidence show?

Some of the strongest mechanistic clues come from experimental models rather than human clinical trials. In 2023, Seo and colleagues reported that the progression of neurodegeneration in a mouse model of tauopathy was influenced by both ApoE genotype and the gut microbiota, providing experimental evidence that microbial context can modify aspects of neurodegenerative pathology.[3]

In 2026, Brossaud and colleagues reported another striking result in a mouse model of Alzheimer’s disease. The short-chain fatty acid butyrate reduced gut and brain amyloid-beta pathology, neuroinflammation and memory deficits in the SAMP8 model. The study also found evidence of early gut abnormalities before memory loss and used human intestinal organoid and colon models to investigate how amyloid beta could affect enteric neuronal connectivity.[4] The findings are biologically compelling, but they remain preclinical evidence. A mouse response to butyrate is not proof that increasing butyrate in humans will prevent dementia.

The prevention problem: biology is only half the challenge

Even if dietary or microbiome changes influence brain health, prevention still depends on whether people can maintain those behaviours over many years. The FINGER trial provides an important comparison. Rather than targeting one biological mechanism, it combines dietary advice, exercise, cognitive training and vascular-risk monitoring. In older adults at risk, the two-year intervention improved cognitive performance compared with general health advice.[5] This supports a broader lesson: dementia risk is multifactorial, so prevention is unlikely to depend on a single food, microbe or supplement.

So, can we rewire the gut?

The most defensible conclusion is that the gut-brain axis is a promising research frontier, not an established Alzheimer’s treatment. Human studies increasingly identify associations between microbiome features and cognitive health, while animal and experimental studies provide plausible mechanisms linking microbial metabolites, gut pathology, inflammation and neurodegeneration.[2-4]

The unresolved question is causality. Which microbial changes matter? When in the disease process do they matter? And can altering them produce a durable improvement in human cognition? Answering those questions will require well-controlled, long-term clinical trials that measure both microbiome biology and meaningful cognitive outcomes.

The future of dementia prevention may therefore be less about choosing between the brain and the gut than understanding how the two interact. The microbiome may not become a hidden cure for Alzheimer’s. But it could become one piece of a much earlier, whole-body strategy for protecting brain health across ageing.

The proposed gut-brain pathway

Diet Fibre-rich foods	Microbiome Microbial metabolism	Metabolites SCFAs such as butyrate	Host Immune + barrier signalling	Brain Neuroinflammation & cognition
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Figure 1. Simplified conceptual model of how dietary patterns may influence the gut-brain axis. This represents a proposed mechanism under investigation rather than an established causal pathway.

Selected references

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